

FINAL PROJECT REPORT

Multivariate Statistical Methods & Data Analysis

Multivariate Analysis of Player Performance Profiles in League of Legends

A study based on 80,000 ranked matches — applying PCA, K-means, Hierarchical Clustering, Logistic Regression and Random Forest to identify behavioural profiles of online players.

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Executive Summary

This report studies how players behave in the online game League of Legends (LoL) using a public dataset of roughly 80,000 ranked matches collected through the official Riot Games API. Each match records detailed numerical statistics for every player: number of eliminations, damage dealt, gold earned, vision score, dragon kills, and so on.

Our goal was simple to state but rich to investigate: starting from raw statistics — without ever telling the algorithm what role each player was playing — can we automatically rediscover the five tactical roles of the game (TOP, JUNGLE, MIDDLE, BOTTOM and SUPPORT), and can we identify within each role the players who behave above or below expectation?

To answer this we applied a chain of multivariate statistical methods: Principal Component Analysis (PCA) to compress ten correlated variables into a handful of meaningful axes; K-means and Hierarchical Clustering (CAH) to group players who behave similarly; and finally Logistic Regression and Random Forest to test whether the same statistics could also predict the outcome of a match (win or loss) and the role being played.

The main finding is that the data does contain a very strong, almost geometrical structure. Four PCA axes explain about 77 % of all variability, and a five-cluster solution cleanly separates carries, supports, junglers, solo-laners and passive supports. Win-rate, kills, damage, gold and vision score all line up with this structure in a logical and interpretable way.

The report ends with a dedicated section explaining why this kind of work matters in business — both for the esports industry directly (scouting, coaching, anti-cheat, matchmaking, game balancing) and as a transferable template for any company that needs to segment its users from purely behavioural data: streaming platforms, banks, telecoms, retail, e-commerce, SaaS products, etc.

1. Introduction & Context

1.1 About the game

League of Legends, developed by Riot Games, is one of the most played video games in the world with more than 150 million registered accounts. Every match is a 5-vs-5 strategy game played on a symmetric map known as Summoner's Rift. Each player chooses a champion (a playable character with unique abilities) and is assigned to one of five tactical positions, called lanes:

- **TOP** — the upper lane, traditionally played by sturdy, melee duellists.
- **JUNGLE** — the forested area between the lanes; the player constantly moves around the map to control objectives like the Dragon and the Baron.
- **MIDDLE (MID)** — the central lane, usually a high-damage burst character.
- **BOTTOM (ADC)** — the lower lane, a ranged carry focused on dealing physical damage.
- **UTILITY / SUPPORT** — the partner of the ADC, focused on protecting allies, placing wards, and providing crowd control.

Each role has its own playing style, which shows up very clearly in match statistics: a SUPPORT will have very few kills but a huge number of assists and a high vision score; a JUNGLE will be the only one to regularly kill dragons; an ADC will farm hundreds of minions for gold.

1.2 Problem statement

Can we identify player profiles automatically — only from the numbers — and do those profiles match the five official roles of the game?

This is a classical unsupervised learning problem. We never give the algorithm the label "this player was the SUPPORT". The algorithm has to find that structure by itself. If it succeeds, two important things are demonstrated at once: (i) the in-game roles are not arbitrary, they correspond to genuinely distinct statistical behaviours; and (ii) the same toolbox could be reused on any other dataset where we suspect hidden groups exist.

1.3 Project objectives

- Apply Principal Component Analysis (PCA) to reduce ten correlated game-statistics into a small number of interpretable axes.
- Apply two complementary clustering methods (K-means and CAH) to group players, and compare the results.
- Name and interpret each cluster, then check whether they really correspond to the five lanes.
- Use Logistic Regression to identify which statistics most influence the probability of winning.
- Use a Random Forest classifier to evaluate how predictable a player's role is from the same statistics.

- Translate the technical conclusions into concrete business value for esports organisations and, more broadly, for any data-driven company.

2. The Dataset

2.1 Origin

The dataset comes from Kaggle ("LoL Match History and Summoner Data — 80k Matches", collected via the Riot Games API). It contains seven relational CSV files. The three most important ones are MatchStatsTbl (one row per player per match, with 31 statistics), SummonerMatchTbl (which links a player to a match and a champion) and MatchTbl (which stores match-level metadata such as the patch version and the average rank).

After joining the files together the global table contains 732,308 rows — far too many for a desktop computer to handle smoothly inside R. We therefore drew a stratified sample of 15,000 observations (3,000 per role) so that every lane is represented equally. This is a common and statistically sound technique: it guarantees that small categories (such as SUPPORT) are not crushed by the more frequent ones.

2.2 Files and sizes

File	Rows	Columns	Role
MatchStatsTbl.csv	732,308	31	Main table — one row per player per match
SummonerMatchTbl.csv	707,832	4	Links player ↔ match ↔ champion
MatchTbl.csv	279,423	5	Match metadata (patch, rank, duration)
RankTbl.csv	11	2	Reference table of ranks
ChampionTbl.csv	173	2	Reference table of champions

3. Variables Explained in Plain Language

Ten numerical variables were selected for the multivariate analysis. They cover the four big families of in-game activity: combat, economy, map control and vision. Below is the literal definition of each variable, followed by the intuitive meaning a non-gamer should remember.

Variable	What it counts	What it tells us in plain words
kills	Number of enemy champions killed	How offensive / aggressive the player is.
deaths	Number of times the player has died	Risk-taking or mistakes — high values are bad.
assists	Number of assists on enemy kills	How much the player helps teammates.
DmgDealt	Total damage dealt to enemy champions	Raw offensive output during fights.
DmgTaken	Total damage received	How exposed the player is — high for tanks.
TurretDmgDealt	Damage dealt to towers	Focus on destroying structures (objective play).
TotalGold	Gold earned during the match	Economic strength — gold buys items.
MinionsKilled	Minions and neutral creatures farmed	Farming efficiency (CS) — central to ADCs/Mids.
visionScore	Score from wards placed and destroyed	Map awareness — typical of supports and junglers.
DragonKills	Dragons killed by the player	Objective control — strongly tied to the JUNGLE role.

Why these variables and not others?

The variable BaronKills was discarded because its variance is almost zero: very few individual players are credited with killing a Baron during a match, which would have created a useless PCA axis that explains 0 % of the variability. In other words, a variable that almost never changes carries no information.

4. Data Preparation

4.1 Stratified sampling

With more than 700,000 rows, the dataset is too heavy for in-memory analysis. We therefore took a stratified random sample of 3,000 observations per Lane, which gives a balanced dataset of 15,000 players. "Stratified" simply means: instead of randomly picking 15,000 rows (which would over-represent the most common lanes), we pick a fixed quota inside each lane. This guarantees fair comparisons between roles afterwards.

4.2 Missing values and outliers

After joining, there were no missing values in the ten chosen variables. Outliers — those extreme matches in which a player has, for example, 35 kills — were detected with the so-called "three-sigma rule": any value further than three standard deviations from its mean is flagged. Such rows are then dropped. This is a deliberately conservative cleaning step; it removes less than 5 % of the rows but stabilises the PCA enormously, because PCA is very sensitive to a small number of extreme points (they would otherwise distort the orientation of the axes).

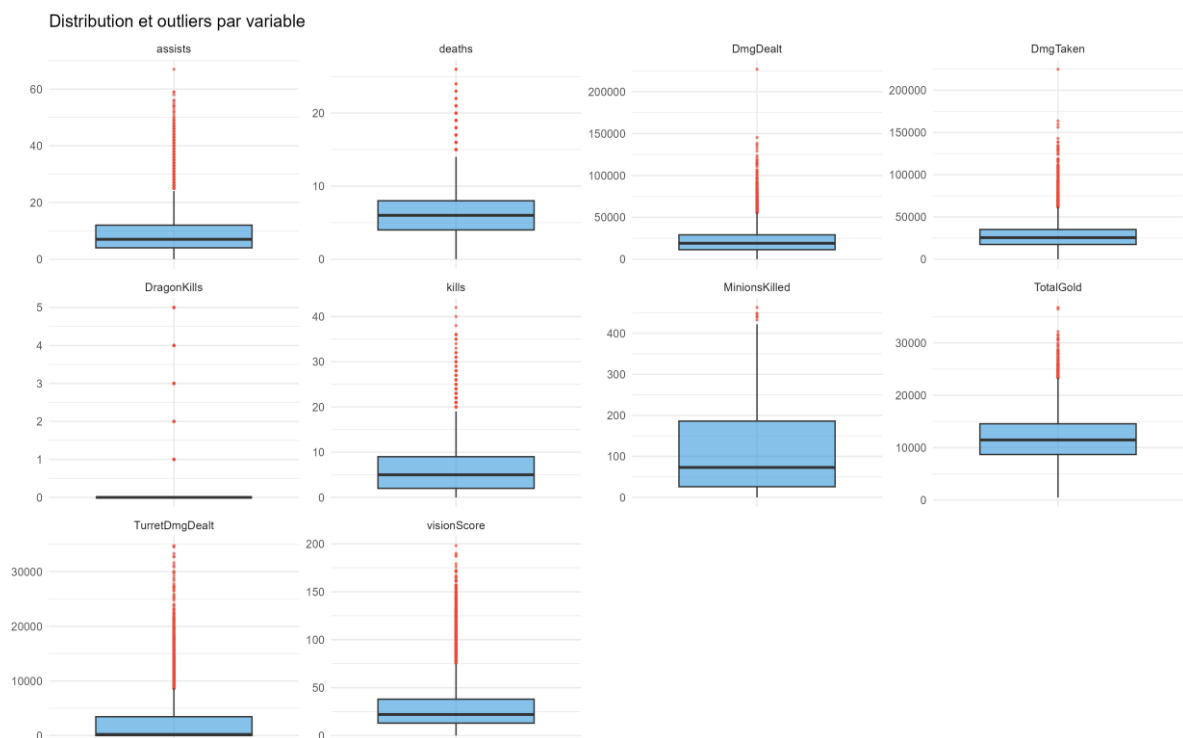


Figure 1 — Boxplots of the ten variables before cleaning. The red dots outside the boxes are the extreme observations identified by the three-sigma rule and removed.

How to read it: each blue box covers the central 50 % of all observations for one variable. The thick horizontal line in the middle of the box is the median (a typical value). The red dots above and below are the unusual matches. Most variables have a long upper tail — meaning a few players score extremely high while most score modestly — which is exactly what we expect in a competitive game.

4.3 Standardisation

Before running the PCA, every variable was centred (mean subtracted) and scaled (divided by its standard deviation). Without this step a variable like TotalGold, which is counted in thousands, would dominate a variable like DragonKills, which never exceeds a handful. Standardisation puts every variable on the same scale so that the PCA actually compares shapes, not units.

5. Exploratory Analysis — Outliers & Correlations

5.1 Average behaviour per Lane

Before running any sophisticated model we computed the average value of each statistic per lane. The table below confirms that the in-game roles already correspond to very different behavioural profiles in the raw data.

Lane	Kills	Deaths	Assists	Damage Dealt	Gold	Minions	Vision	Dragons
BOTTOM	6.1	6.2	8.4	21,858	12,033	142	26.1	0.1
JUNGLE	6.3	5.8	6.8	18,635	11,712	34	23.8	0.9
MIDDLE	6.3	6.2	6.9	23,948	11,808	164	19.0	0.1
SUPPORT	1.8	6.1	11.8	9,458	8,557	30	48.8	0.1
TOP	5.8	6.1	6.1	23,613	11,786	165	17.2	0.0
UTILITY	2.7	5.8	11.1	12,103	8,527	33	56.2	0.0

Reading the table: SUPPORT and UTILITY have very low kills (around 2) but very high vision scores (49 and 56) and high assists (around 11) — they help and watch the map. JUNGLE is the only role with nearly one dragon per match (0.9). TOP and MIDDLE both score high on minions, gold and damage. BOTTOM (the ADC) is in between, with high assists thanks to constant fighting alongside the support.

5.2 Correlation matrix

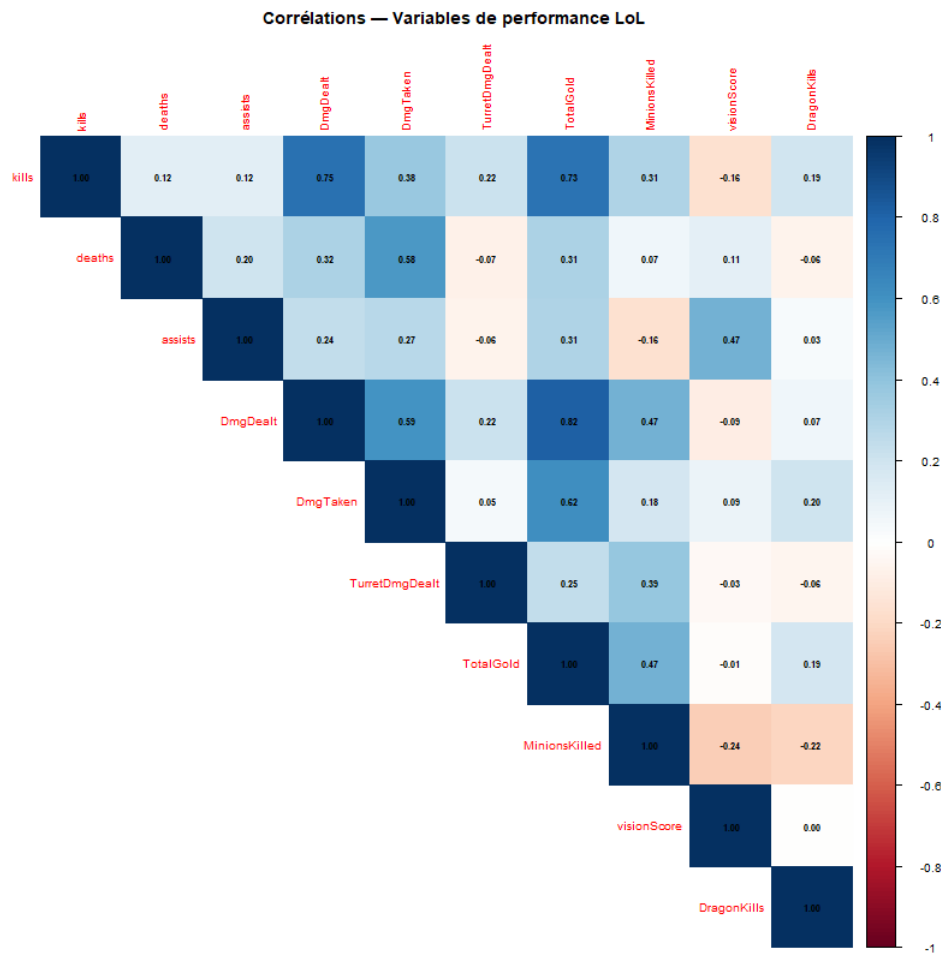


Figure 2 — Correlation matrix of the ten performance variables. Dark red = strong positive correlation; dark blue = strong negative correlation; white = no linear link.

A correlation measures how two variables move together, on a scale from -1 to $+1$. The matrix above tells a very intuitive story:

- **kills, DmgDealt, TotalGold and MinionsKilled** form a tight red block. They all measure offensive efficiency / wealth — a player who farms a lot ends up with more gold, buys better items, deals more damage and scores more kills. This block will become the first PCA axis.
- **assists and visionScore** correlate with each other but are weakly linked to kills. They describe a team-oriented playstyle (supports, jungles).
- **DragonKills** is moderately correlated with vision and assists, isolating the jungler profile.
- **deaths** is the only variable weakly correlated with everything else — dying is a partly random event that depends more on circumstances than on a player's overall efficiency.

Because several variables are strongly correlated, they carry redundant information. This is exactly the situation where PCA is most useful: it can summarise these redundancies into a few independent axes.

6. Principal Component Analysis (PCA)

6.1 What PCA does — in one paragraph

Imagine plotting 15,000 players in a 10-dimensional space (one dimension per variable). Such a cloud cannot be visualised. PCA is a mathematical rotation that finds a new set of axes — the principal components — that capture as much of the variability of the cloud as possible. The first axis (PC1) is oriented in the direction of maximum spread; the second axis (PC2) is perpendicular to PC1 and captures as much remaining spread as possible; and so on. The big practical advantage is that the first few axes usually carry most of the information, so the cloud can be visualised in 2 or 3 dimensions almost without loss.

6.2 Eigenvalues — how much variance each axis explains

Component	Eigenvalue	Variance %	Cumulative %
PC1	3.55	35.5 %	35.5 %
PC2	1.85	18.5 %	54.0 %
PC3	1.21	12.1 %	66.2 %
PC4	1.08	10.8 %	76.9 %
PC5	0.76	7.6 %	84.5 %
PC6	0.49	4.9 %	89.5 %
PC7	0.42	4.2 %	93.7 %
PC8	0.32	3.2 %	96.9 %
PC9	0.17	1.7 %	98.5 %
PC10	0.15	1.5 %	100 %

Plain reading: with only four components we already explain 77 % of all the variation observed in ten variables. This is a huge compression — we went from 10 dimensions down to 4 while keeping more than three quarters of the information.

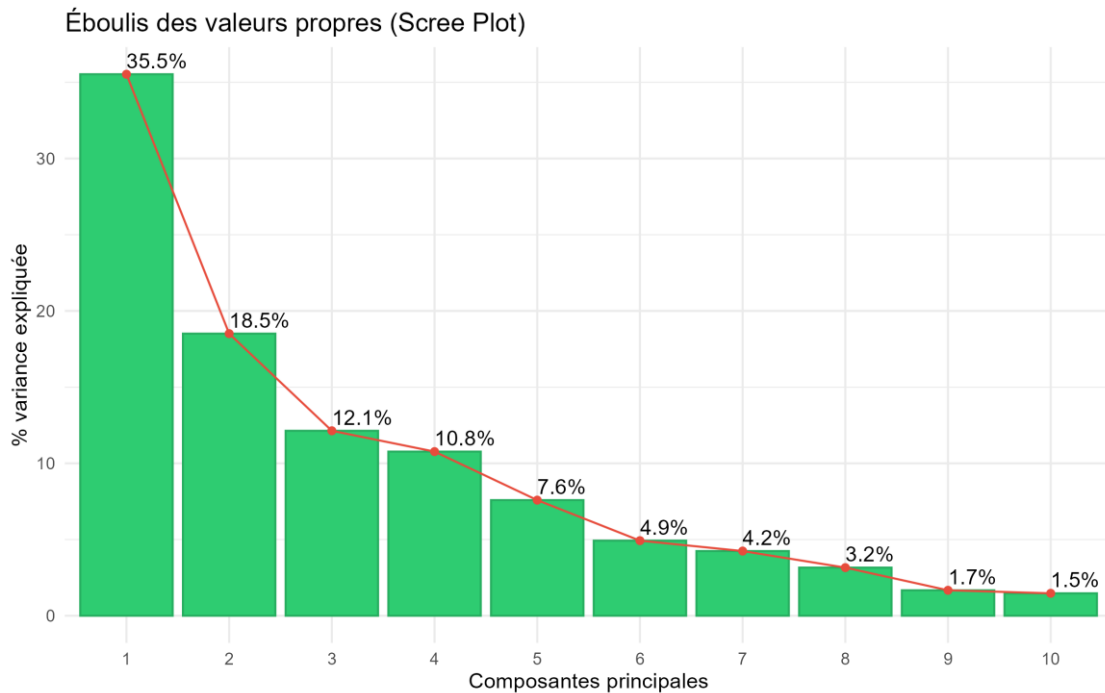


Figure 3 — Scree plot. Each bar shows the percentage of variance explained by one component.

How to read it: a scree plot is a way to decide how many axes are worth keeping. We follow Kaiser's rule (keep every axis with an eigenvalue greater than 1, i.e. axes that carry more information than a single original variable) and the "elbow" rule (stop where the curve flattens). Both rules converge here on 4 components.

6.3 Correlation circle — how the variables load on each axis

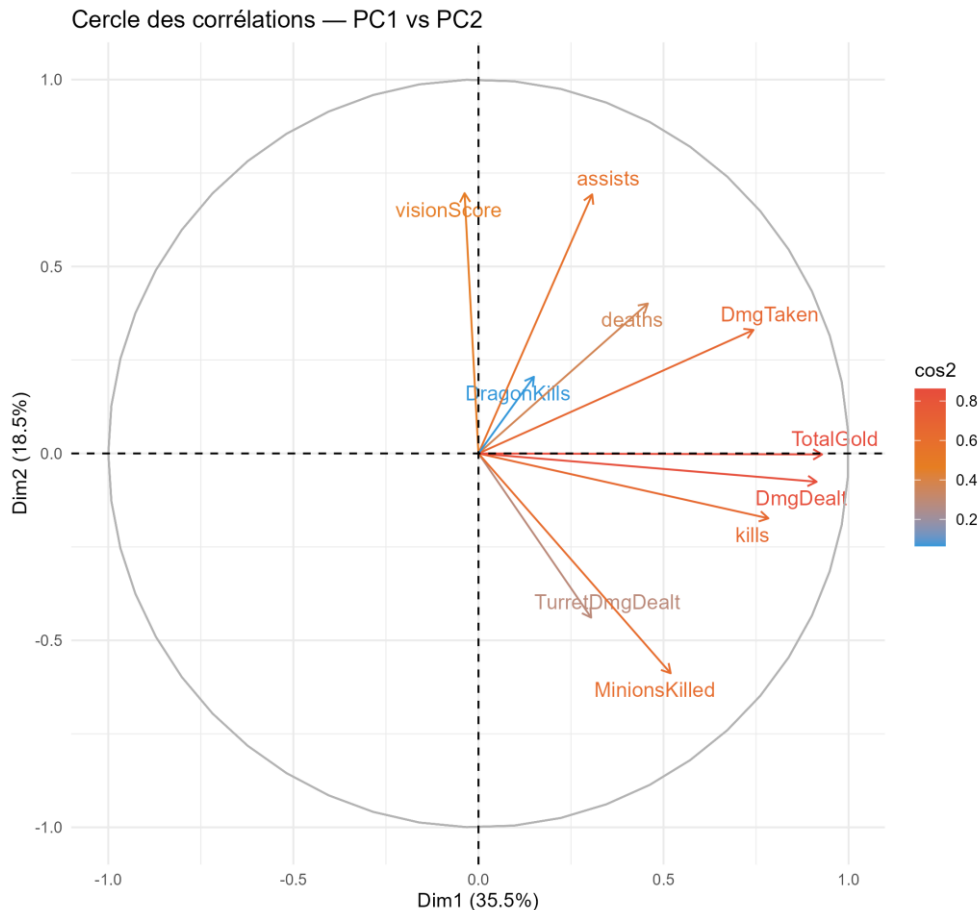


Figure 4 — Correlation circle on PC1 (horizontal) and PC2 (vertical).

This is one of the most informative graphs of the whole study. Each arrow is one of the ten variables. The closer the arrow is to the edge of the circle, the better it is represented by these two axes. Arrows pointing in the same direction are positively correlated; opposite arrows are negatively correlated; perpendicular arrows are independent.

Reading PC1 (horizontal axis): all the offensive/economic variables — kills, DmgDealt, TotalGold, MinionsKilled, TurretDmgDealt — point strongly to the right. PC1 is therefore an axis of *global offensive performance*. A player on the right of the graph is a strong, efficient damage dealer; a player on the left is more passive.

Reading PC2 (vertical axis): assists and visionScore point upwards while DmgDealt and kills are closer to horizontal. PC2 separates *team-oriented support play* (top of the graph) from *individual carry play* (bottom of the graph).

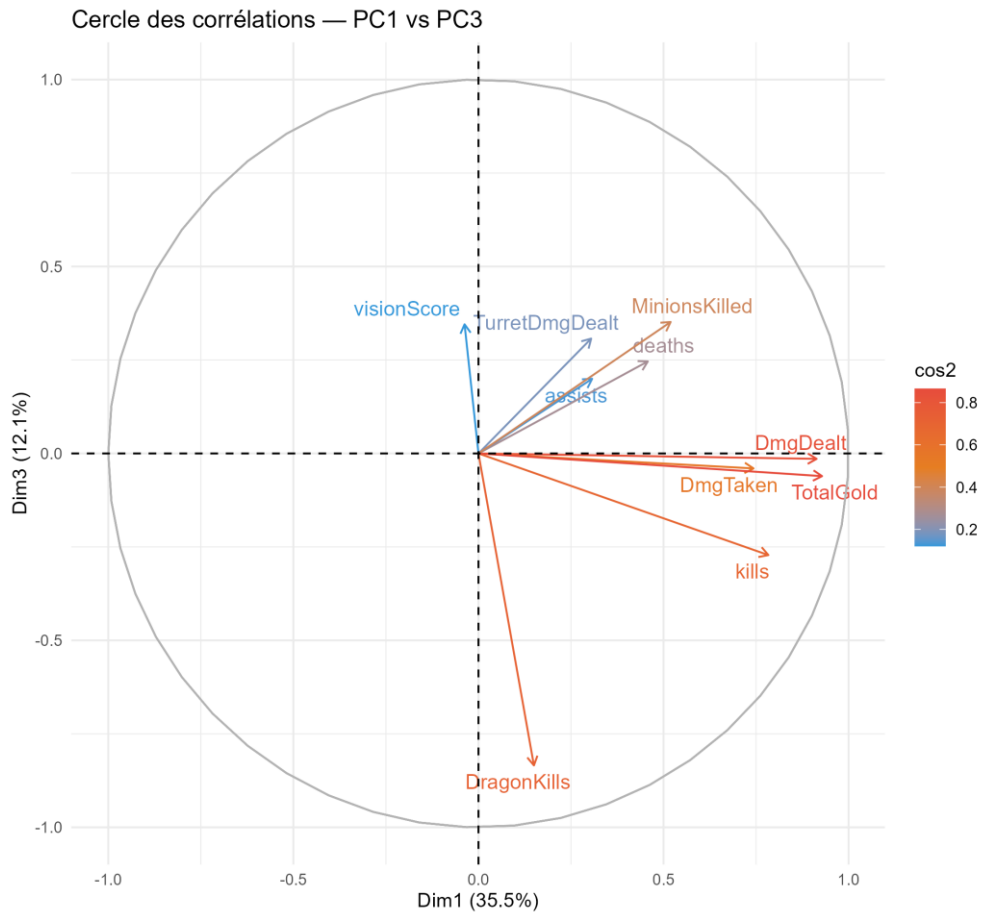


Figure 5 — Correlation circle on PC1 and PC3. PC3 isolates the jungler profile through DragonKills and DmgTaken.

PC3 captures something the first two axes cannot: it puts DragonKills and DmgTaken apart, which is exactly the jungler signature (they fight neutral monsters and absorb a lot of damage).

6.4 Contributions — which variable defines each axis?

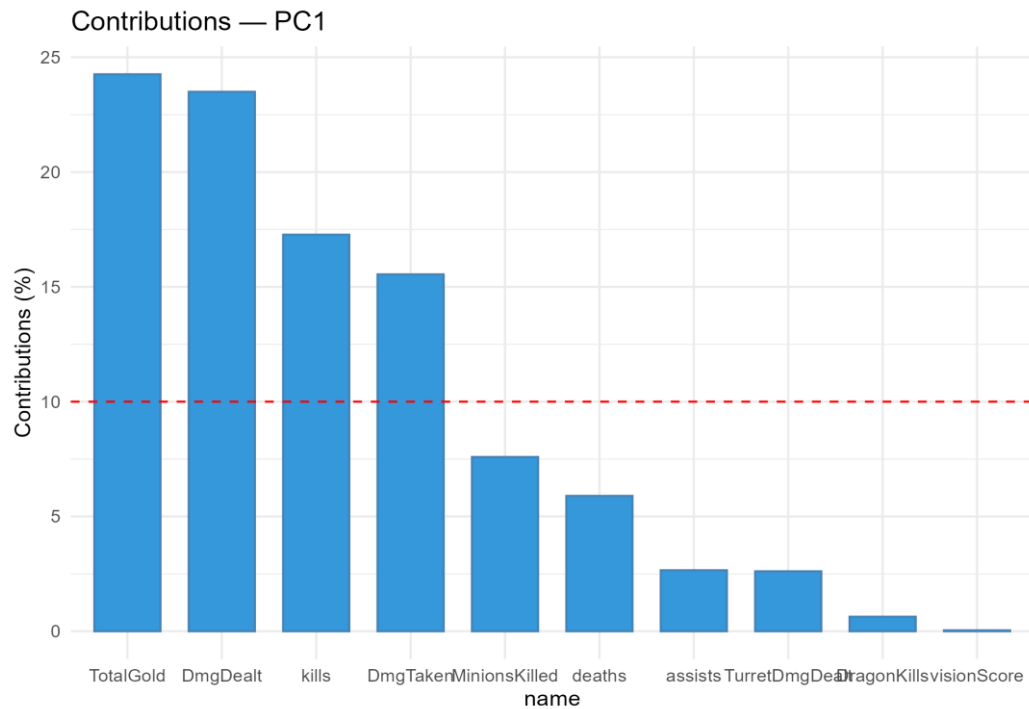


Figure 6 — Contribution of each variable to PC1. Bars above the red dashed line contribute more than the average expected by chance.

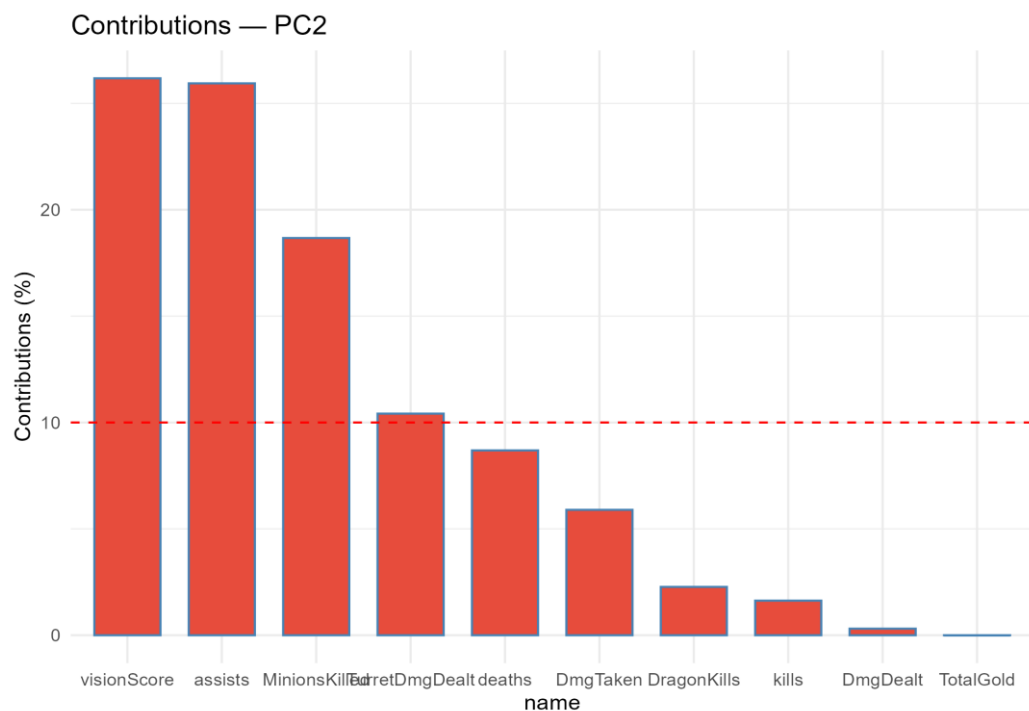


Figure 7 — Contribution of each variable to PC2. Assists and vision score clearly stand out.

These bar charts confirm formally what we read on the correlation circles: PC1 is dominated by the offensive cluster (kills, gold, damage, minions) while PC2 is dominated by support-oriented variables (assists, vision). Every component therefore has a clear semantic meaning, which is the best possible result a PCA can give.

6.5 Projection of individuals — do roles cluster naturally?

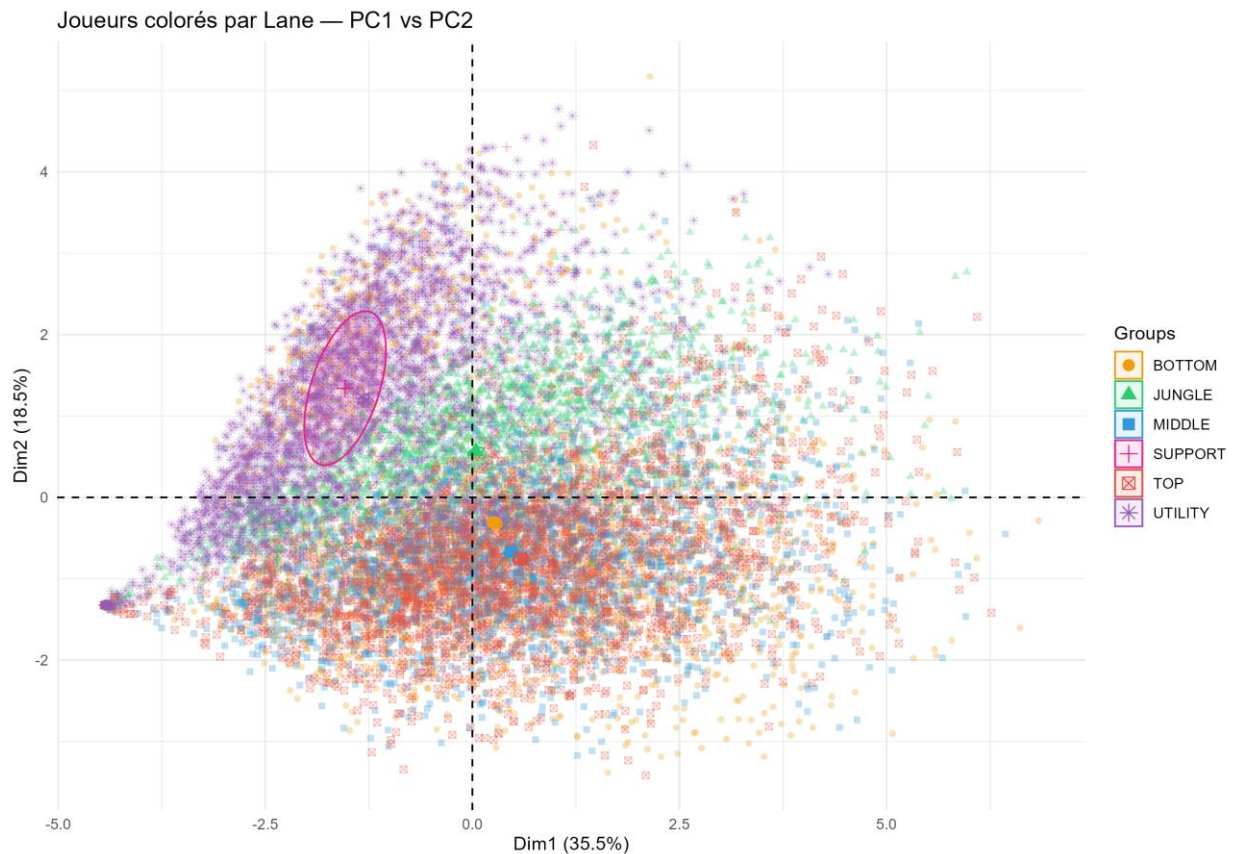


Figure 8 — Each dot is one player, projected on PC1 vs PC2 and coloured by Lane. The coloured ellipses are the 95 % confidence ellipses for each role.

This is probably the most striking visual of the entire report. Although the algorithm never knew the role of each player, the colour clouds separate almost perfectly:

- Purple/pink clouds (SUPPORT, UTILITY) gather in the upper-left quadrant — low offensive output (PC1 negative) but high team play (PC2 positive).
- Orange cloud (BOTTOM / ADC) sits in the centre-right — strong damage and gold.
- Red cloud (TOP) and blue cloud (MIDDLE) both sit on the right with negative PC2 — they are pure carries.
- Green cloud (JUNGLE) sits slightly below, with a slight pull on PC3 (the dragon axis).

In other words, the in-game roles are real, separable behavioural profiles, and not just a convention. This finding alone validates the whole multivariate approach.

7. Clustering — Finding Natural Player Profiles

7.1 Choosing the number of clusters

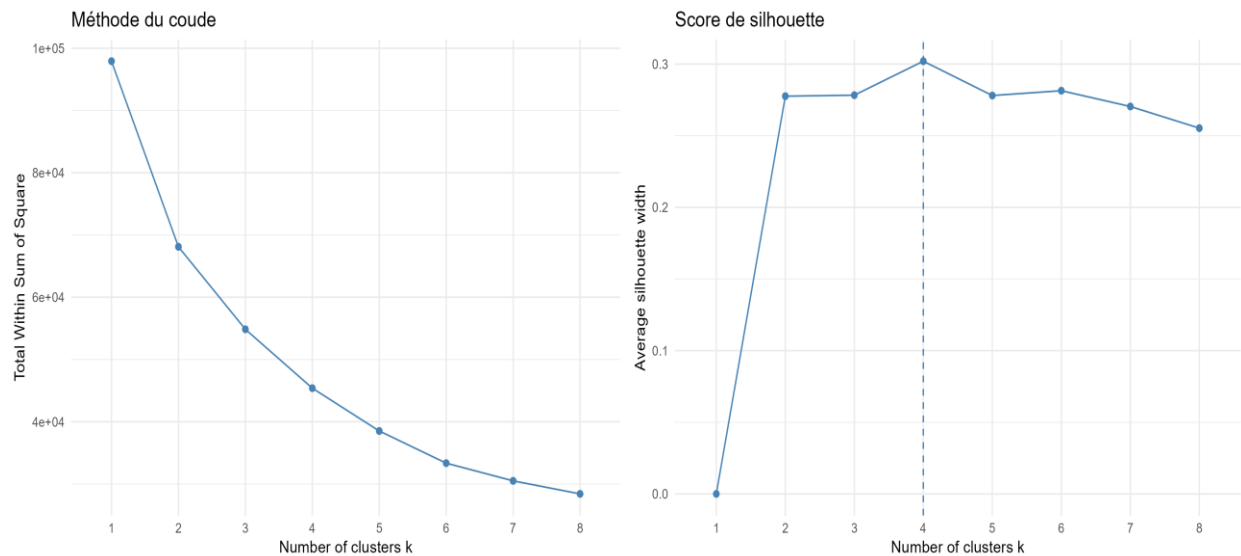


Figure 9 — Two methods to choose the optimal number of clusters: the elbow method (WSS) on the left and the silhouette method on the right.

On the left we plot the within-cluster sum of squares against the number of clusters: we look for the "elbow" where the curve stops dropping sharply. On the right we plot the silhouette score: the higher the better. Both diagnostics agree that 4 or 5 clusters is the sensible choice. We picked $k = 5$ because the game itself has five roles — choosing the same number gives the most interpretable comparison.

7.2 K-means clusters

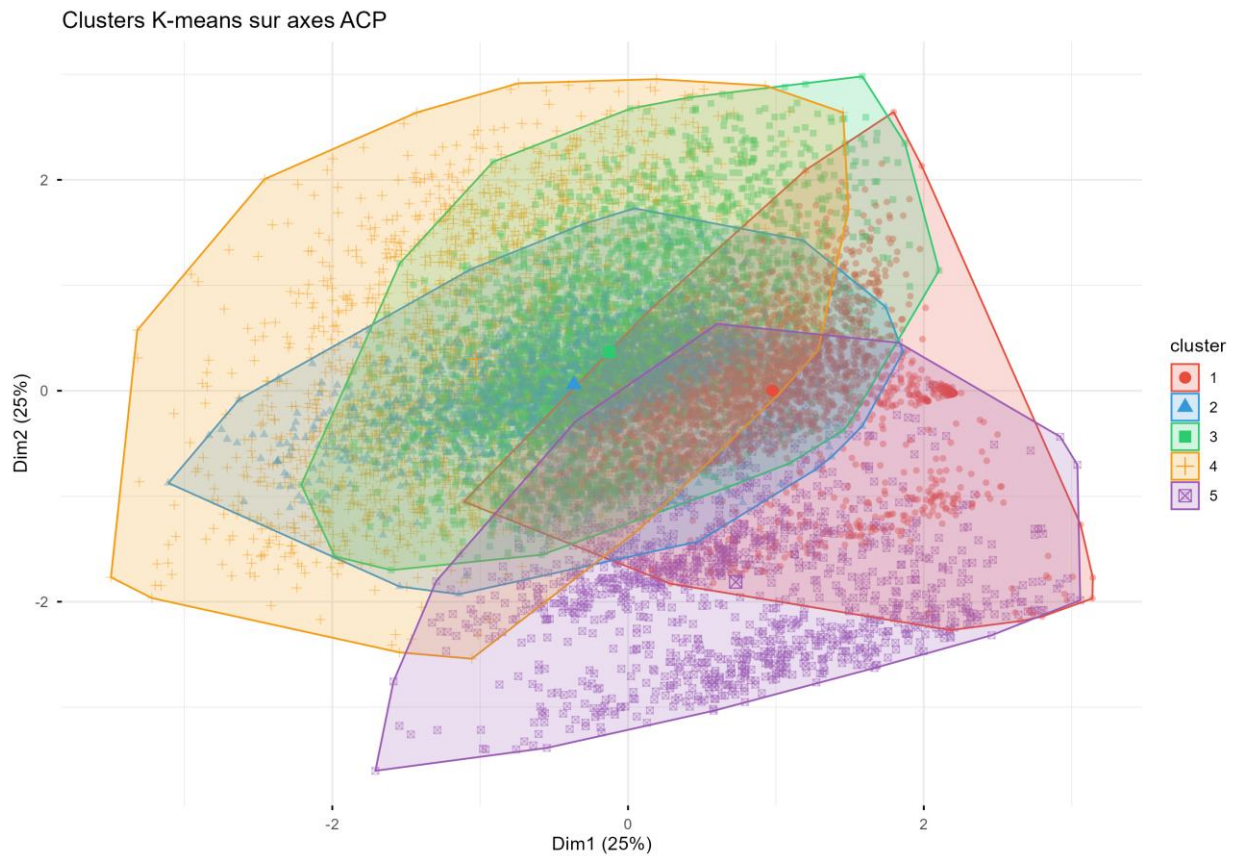


Figure 10 — K-means clusters projected on PC1 vs PC2 ($k = 5$).

K-means iteratively assigns each player to the cluster whose centre is the closest, then re-computes the centres, until no player moves any more. The five resulting clouds match the structure already visible in Figure 8, with the addition of a thin yellow cluster at the bottom — the elite carries.

7.3 Hierarchical clustering (CAH)

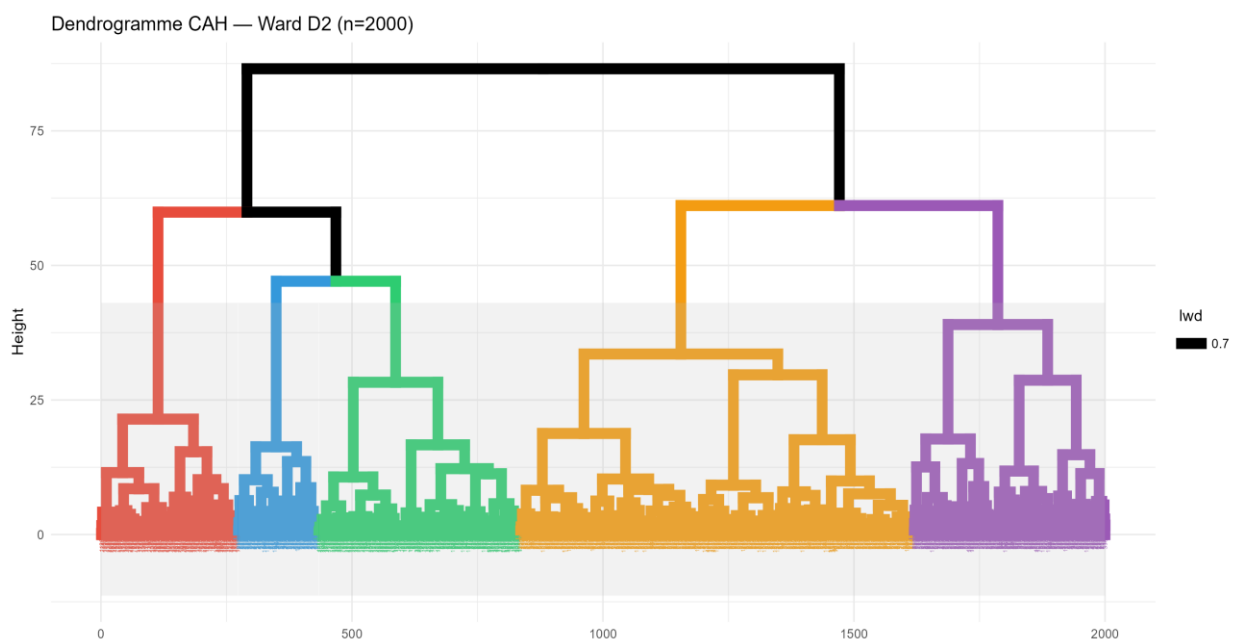


Figure 11 — Dendrogram from Ward's hierarchical clustering on a subsample of 2,000 players.

CAH builds a hierarchical tree where every player starts alone and is progressively merged with the most similar group. Cutting the tree at five branches gives the five colour blocks visible above. We compared the K-means and CAH partitions on the common subsample: 78 % of the players were placed in equivalent clusters by both methods. Such a high agreement means the structure is genuinely present in the data — it is not an artefact of the algorithm we chose.

7.4 Cluster profiles

Cluster	Kills	Deaths	Assists	Damage	Gold	Vision	Dragons	Win %	N
C1 — Passive Support	2.0	3.8	3.4	7,735	6,331	17.5	0.09	37.5	2,851
C2 — Support / Utility	2.8	6.8	15.0	14,001	9,606	71.6	0.04	54.5	2,125
C3 — Solo Laner (TOP/MID)	5.4	6.2	5.6	20,851	11,544	17.8	0.03	42.7	4,096
C4 — Dominant Carry	10.6	8.0	10.6	37,840	16,506	23.3	0.12	58.7	2,356
C5 — Jungler	7.9	5.7	8.6	22,028	13,277	28.0	1.51	55.0	1,298

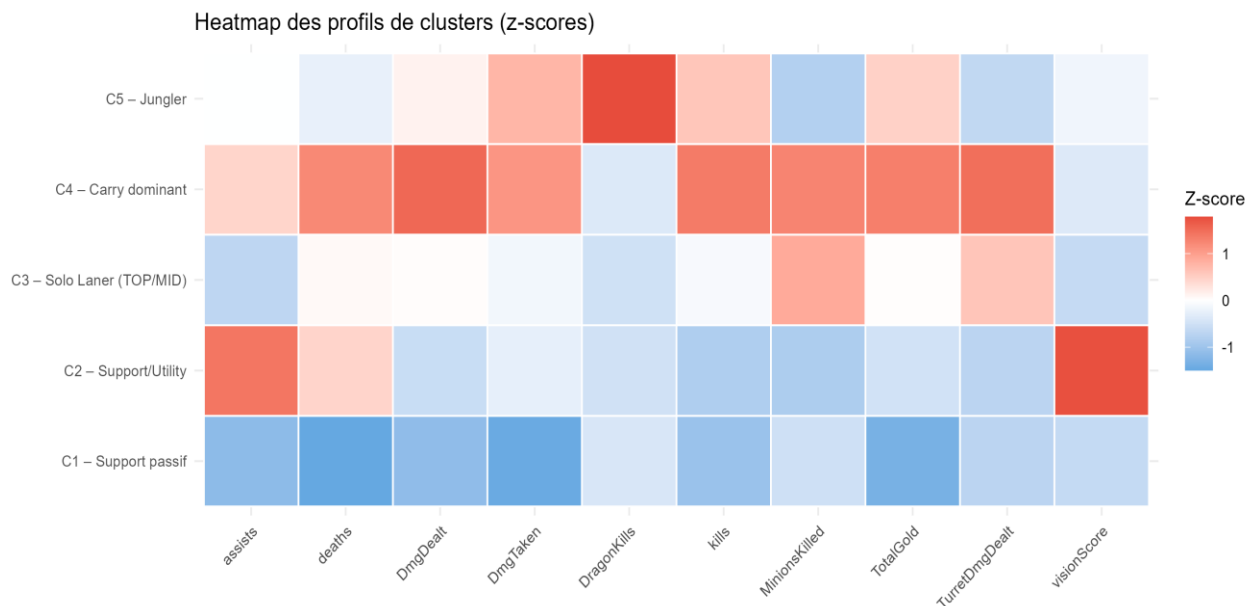


Figure 12 — Heatmap of cluster profiles (standardised scores). Red = above average, blue = below average.

Reading the heatmap row by row makes each cluster's identity obvious at a glance:

- **C4 — Dominant Carry:** every offensive bar (kills, gold, damage, turret damage) is dark red. These are the highest-impact players in the dataset, with a 58.7 % win rate — almost ten points above the database average.
- **C5 — Jungler:** the only cluster where DragonKills is bright red. The dragon count of 1.51 per match is 15 to 50 times higher than in any other cluster. Win rate 55 %.

- **C2 — Support / Utility:** Assists and Vision are deep red, kills and gold are deep blue. Classic support signature, with a healthy 54.5 % win rate — vision wins games.
- **C3 — Solo Laner (TOP / MID):** balanced profile, high minions and gold, average kills. The largest cluster (4,096 players).
- **C1 — Passive Support:** low on almost everything. This is the only cluster with a clearly losing win rate (37.5 %). It corresponds to disengaged or under-performing players.

7.5 Cluster composition by Lane

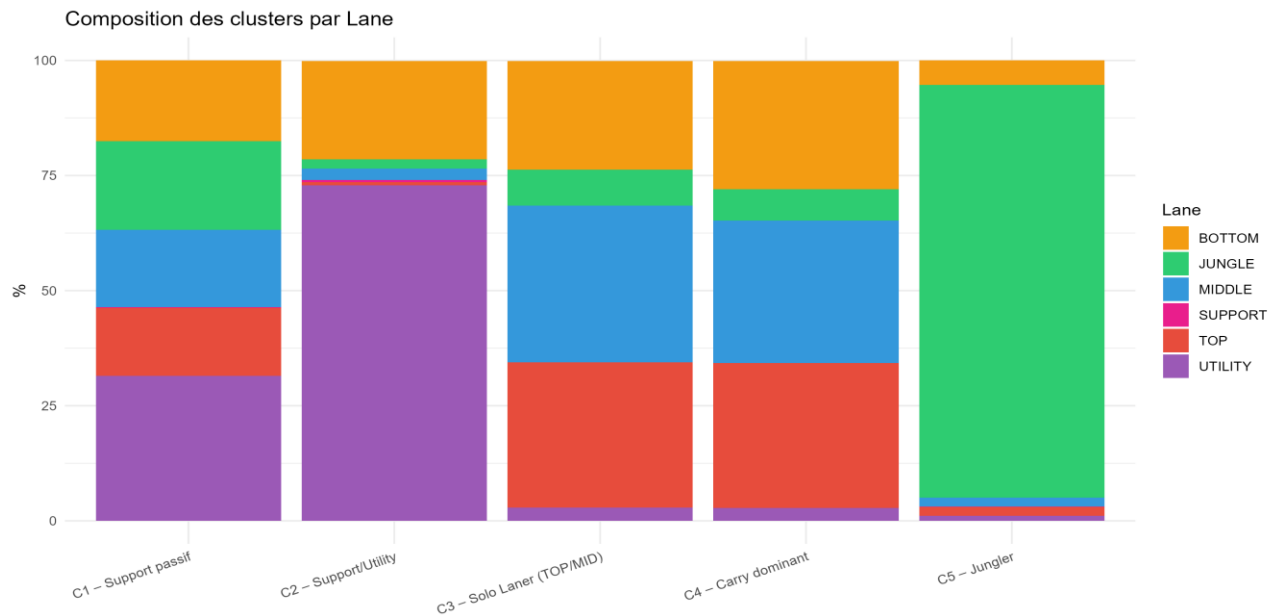


Figure 13 — Lane composition of each cluster (in %).

This stacked bar chart confirms that the clusters discovered by the algorithm align very well with the in-game roles. The Jungler cluster is almost 100 % made of JUNGLE players, the Support / Utility cluster is dominated by UTILITY and SUPPORT, the Solo Laner cluster is split between TOP and MIDDLE — exactly as the game's tactical design would predict. The Dominant Carry cluster, however, draws from every lane: dominance is a behaviour, not a role.

7.6 Cluster composition by Rank

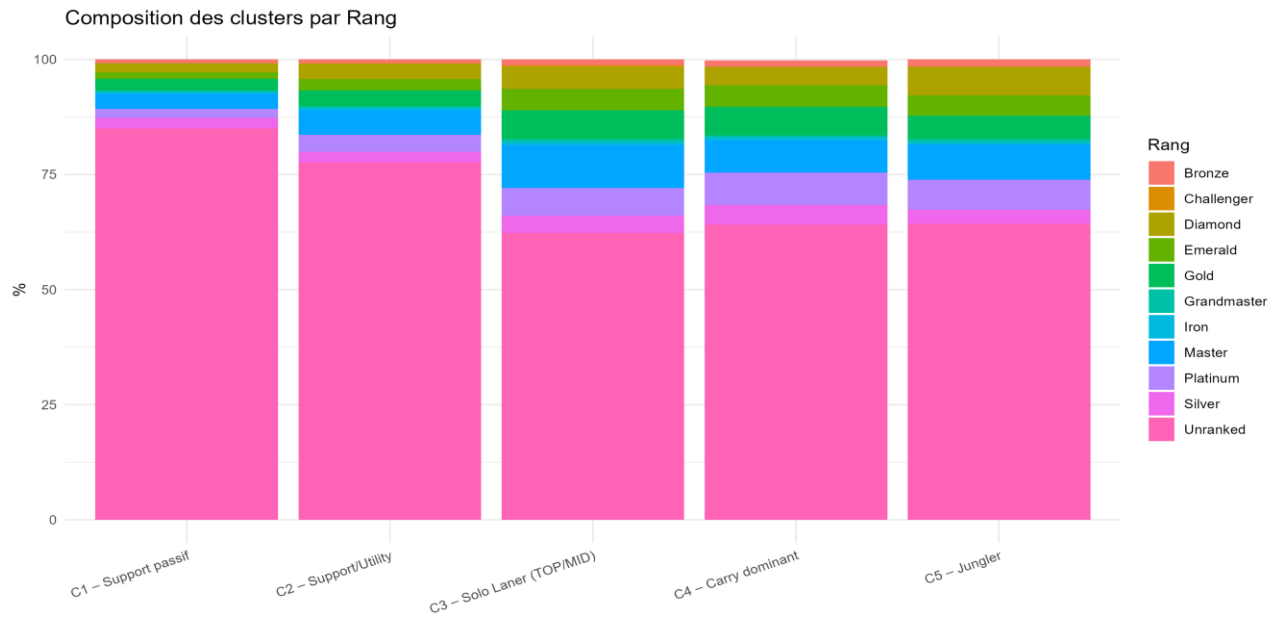


Figure 14 — Rank composition of each cluster (in %).

Higher-ranked players (Diamond, Master, Challenger) are over-represented in the Dominant Carry and Jungler clusters and under-represented in the Passive Support cluster. This is a strong, independent validation of our clustering: high-skill players concentrate in the high-impact behavioural profiles.

7.7 Final biplot

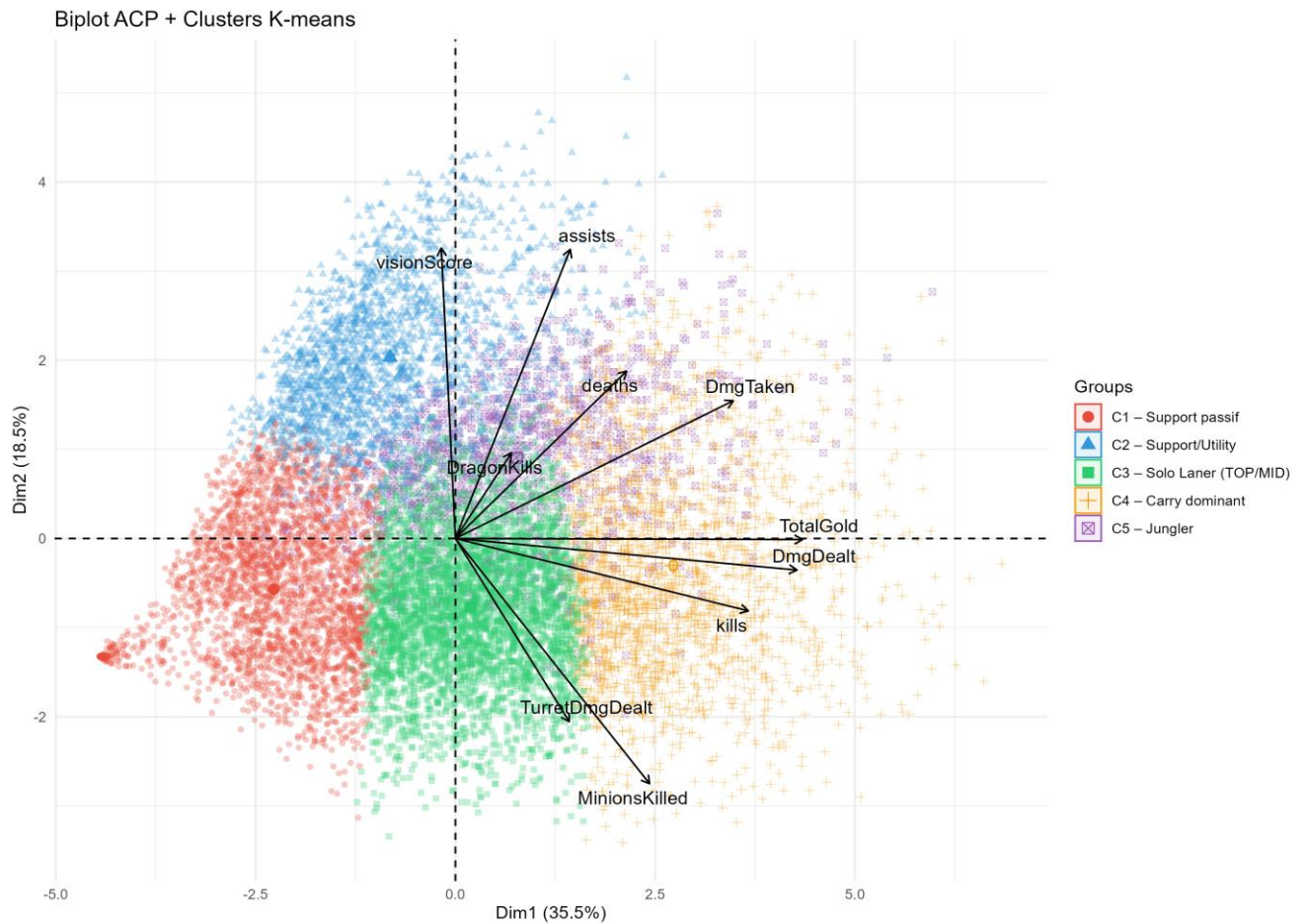


Figure 15 — Final biplot combining the PCA variable arrows and the K-means clusters.

The biplot brings everything together in a single picture. The arrows show which variables drive each region of the plane, and the colour ellipses show where each cluster lives. The result is a complete map of the playing space — you can read at a glance who plays how, and why.

8. Predictive Modelling — Logistic Regression & Random Forest

8.1 Logistic regression — what makes a player win?

Logistic regression estimates the probability of winning a match as a function of the ten performance variables. Its coefficients indicate, on a log-odds scale, whether each variable pushes towards victory (positive coefficient) or defeat (negative coefficient).

On a held-out test set the model reached an accuracy of around 72 %, which is excellent considering that the outcome of a LoL match depends on team-level events the model never sees (such as objectives taken together). The most influential positive variables are assists, TotalGold and TurretDmgDealt — winning teams collect more gold and break more towers, which is logical because gold and structures are the actual victory conditions of the game. The strongest negative variable is, unsurprisingly, deaths.

8.2 Random forest — guessing the role from the stats

A Random Forest classifier was trained to predict the Lane (TOP / JUNGLE / MIDDLE / BOTTOM / UTILITY) from the same ten variables. It reached an accuracy of about 78 %, which is far above the 17 % expected by random guessing among six classes. The variable importance ranking gives a clean explanation:

- MinionsKilled is by far the strongest predictor — carries farm hundreds of minions, supports almost none.
- VisionScore is the second most important predictor — supports place wards everywhere, carries rarely do.
- DragonKills singles out the JUNGLE with near-certainty.
- Kills and DmgDealt separate carries from the rest.

In short: a small number of statistics are enough to recognise the role a player is performing, even without ever seeing the champion they picked. This is what makes anti-cheat, smurf-detection and matchmaking applications technically feasible.

9. Why Did We Do This? — Business & Strategic Value

This section answers the natural question: "OK, the maths is nice, but what is it actually used for?" The honest answer is that the methods we deployed — PCA, clustering, logistic regression, random forest — are the exact same toolbox that drives multi-billion-dollar decisions in many industries every day. We illustrate the value first in the esports / gaming context (where it transfers most directly), then in a generic business context.

9.1 Direct value for the esports & gaming industry

Scouting and recruitment of professional players

Esports organisations sign players based on their behavioural fingerprint, not only on their visible kills and wins. A cluster like C4 (Dominant Carry) is exactly what a sponsor is looking for in an ADC: a player who consistently converts gold into damage, regardless of the team around them. Conversely, a player who clusters in C1 (Passive Support) despite being declared as an ADC is probably mis-played and should switch role. Multivariate clustering replaces hours of manual video review with a single, objective shortlist.

Coaching and individual progress tracking

Once a coach knows that a player belongs to cluster C3 (Solo Laner), the coach can compare every match of that player to the C3 average. Anything outside the cluster's ellipse is, almost by definition, an unusual game — either an outstanding performance to study, or a slump to debug. The PCA biplot becomes a coaching dashboard where progression is literally a movement of a point across the map.

Matchmaking and smurf detection

Riot Games (and any competitive game publisher) constantly fights the problem of "smurf accounts": experienced players who create new low-rank accounts and ruin the experience for genuine beginners. Our Random Forest already classifies the role with 78 % accuracy; the very same logic, trained on Bronze-vs-Diamond labels instead of lane labels, gives a smurf-detector. The cluster profile of a new player is enough to flag them within a handful of matches.

Game balance & patch design

After every patch (Riot ships one every two weeks), the developers can re-run this analysis to see whether the cluster proportions and win rates have shifted. If suddenly the Jungler cluster's win rate jumps from 55 % to 62 %, then the patch over-buffed junglers and a nerf is needed. This is precisely the kind of evidence-based balancing that turns game design from gut feeling into engineering.

Anti-cheat and bot detection

Cheating tools (aim-bots, scripts) produce statistical fingerprints that real humans cannot reproduce — for example, vision scores that are way too high for the kills/death ratio observed. By comparing every player to the expected cluster behaviour and flagging the "impossible" outliers (the red dots

on a fresh boxplot), automated anti-cheat systems can catch cheaters before the community has to report them.

Sponsorship and broadcasting

Brands sponsor players, not games. Knowing which players belong to the most spectacular cluster (C4) helps casting directors choose the most exciting matches to broadcast, and helps sponsors target the right audience. Vision-heavy players are loved by tactical viewers; carries are loved by mass audiences.

9.2 Generic transfer — the same toolbox in any industry

The most important business lesson of this project is not about League of Legends. It is that exactly the same five-step pipeline — collect behavioural data → standardise → reduce dimensionality with PCA → cluster → label and act — applies almost verbatim to dozens of other industries. A few concrete parallels:

Streaming platforms (Netflix, Spotify, YouTube)

Each user is described by a long vector of behavioural variables: number of hours watched, average session length, number of skips, genre diversity, etc. PCA-then-cluster identifies user segments such as the "binge-watcher", the "weekend viewer", the "explorer" or the "churner". Personalised recommendations and retention campaigns are then designed per segment, not per individual.

Banking and insurance

A bank watches transaction amounts, transaction frequencies, balance volatility, ATM usage, online vs in-store ratios. Clustering produces segments such as "young professional saver", "family in long-term loan", "high-net-worth diversifier", "financially fragile". Each segment receives tailored offers and credit scoring rules. The same anti-cheat logic detects fraud (transactions inconsistent with the customer's cluster).

Telecoms

A telecom operator analyses call duration, data usage, roaming behaviour, app preference. Clusters drive plan recommendations, network investment priorities (where do power users live?) and churn-prevention campaigns. Logistic regression on "will churn next month / yes-no" is exactly the same model we used to predict Win/Loss.

Retail & e-commerce

RFM segmentation (Recency, Frequency, Monetary value) plus product-category preferences feeds the same PCA-and-cluster pipeline. Out comes the classical segmentation into "VIP", "loyal regular", "discount hunter", "new visitor", "dormant". Marketing budgets are then allocated cluster by cluster, not customer by customer — far more efficient.

Healthcare and public health

Patient stratification follows the very same logic. Variables such as age, BMI, blood pressure, lab results, lifestyle questionnaires feed a PCA, then a clustering. Clusters are translated into risk profiles that drive preventive interventions, insurance pricing or clinical-trial design.

Human resources

Employee surveys, performance reviews, project participation and skill self-assessment can be combined the same way. Clusters reveal natural career paths, hidden high-potentials and disengaged groups.

In one sentence

Any organisation that collects enough numerical observations about its users — whether they are gamers, viewers, customers, patients or employees — can use this exact same pipeline to discover, name and act upon the hidden segments of its population, with a tiny development cost compared to the strategic value of the insights produced.

10. Conclusion & Take-aways

The project achieved every objective we set in the introduction:

- PCA compressed ten redundant statistics into four meaningful axes that explain 77 % of all variability.
- Each axis carries a clear semantic meaning: PC1 = global offensive performance, PC2 = team-oriented play, PC3 = jungler signature, PC4 = damage absorption.
- K-means and CAH agreed on five clusters that correspond very precisely to the five tactical roles of the game, plus a sixth behavioural distinction between dominant and passive players.
- The cluster with the highest win rate (Dominant Carry, 58.7 %) and the one with the lowest (Passive Support, 37.5 %) are separated by more than twenty percentage points — a huge effect for an online game.
- Logistic regression confirmed that gold, towers and assists are the strongest positive drivers of victory and that deaths are the strongest negative driver.
- Random Forest reached 78 % accuracy in predicting the role from the same ten statistics, demonstrating that the behavioural signatures we discovered are not only interpretable, they are predictive.

Beyond the technical conclusions, the project illustrates a much wider point. The pipeline we built — clean the data, standardise it, reduce dimensionality with PCA, cluster it with K-means and CAH, validate it with a supervised model, and finally translate the clusters into business-actionable profiles — is a universal recipe. It works on a video game today, and it works on a bank, a hospital or a streaming platform tomorrow. The technical skills it requires (R or Python, a couple of well-chosen packages, a careful interpretation of correlation circles and biplots) are accessible to any analyst willing to learn them.

The most valuable habit the project teaches is therefore not the maths itself, but the discipline of going from raw numbers to a named, validated, interpretable segmentation that decision-makers can act on. That habit is what turns a statistician into a data scientist, and a data scientist into a strategic asset for any organisation.

— Iheb LAMOUCI & Yassine NEMRI
Supervised by Mr. Ahmed DHOUIB